

SKILL-3

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Sec-2

1. To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.

To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind.

A screenshot of a terminal window with a dark background. The window title is 'rehan_lenovo@Rehan-Lenovc'. The terminal shows the GNU nano 8.7.1 editor with C code for a command history program. The code includes headers for stdio, stdlib, and string, and defines a MAX constant of 10. The main function uses a while loop to handle user input. It checks for 'exit' and 'history' commands. For 'history', it prints the stored commands. For other commands, it dynamically allocates memory for the history array using malloc and strcpy, then increments the count. Finally, it prints the history and returns 0.

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX 10

int main()
{
    char *history[MAX];
    char command[100];
    int count = 0;

    while (1)
    {
        printf("Rehan$> ");
        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
            break;

        if (strcmp(command, "history") == 0)
        {
            for (int i = 0; i < count; i++)
                printf("%d: %s\n", i + 1, history[i]);

            continue;
        }

        if (count < MAX)
        {
            history[count] = malloc(strlen(command) + 1);
            strcpy(history[count], command);
            count++;
        }

        for (int i = 0; i < count; i++)
            free(history[i]);

        return 0;
    }
}
```

Output:

```

rehan_lenovo@Rehan-Lenovo:~/skill-1$ nano skill3a.c
rehan_lenovo@Rehan-Lenovo:~/skill-1$ gcc skill3a.c -o skill3a
rehan_lenovo@Rehan-Lenovo:~/skill-1$ ./skill3a
Rehan$> ls
Rehan$> pwd
Rehan$> history
1: ls
2: pwd
Rehan$> whoami
Rehan$> histiry
Rehan$> history
1: ls
2: pwd
3: whoami
4: histiry
Rehan$> exit
rehan_lenovo@Rehan-Lenovo:~/skill-1$ |

```

2. To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

```

GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main()
{
    char input[100];
    char *token;

    printf("Enter command: ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    if (strlen(input) == 0)
    {
        printf("Empty command\n");
        return 0;
    }

    token = strtok(input, " ");

    while (token != NULL)
    {
        printf("Token: %s\n", token);
        token = strtok(NULL, " ");
    }

    return 0;
}

```

Output:

```
rehan_lenovo@Rehan-Lenovo:~/skill-1$ nano skill3b.c
rehan_lenovo@Rehan-Lenovo:~/skill-1$ gcc skill3b.c -o skill3b
rehan_lenovo@Rehan-Lenovo:~/skill-1$ ./skill3b
Enter command: ls 3
Token: ls
Token: 3
rehan_lenovo@Rehan-Lenovo:~/skill-1$ |
```